

# SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

*Symbiosis International (Deemed University)*

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**Experiment No.: 2      Name: Vansh Mahant      PRN: 24070521038      Section: A      Batch: A1**

## 1. Experiment Title:

Online Shopping — JavaScript Basics: a web page using HTML, CSS and JavaScript demonstrating primitive data types, explicit type conversion, and a shopping bill calculator that applies conditional discount rates based on the selected payment method.

## 2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, JavaScript

## 3. Experiment Program Code:

[index.html](#)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Online Shopping - JavaScript Basics</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <div class="app-container">
    <!-- Main Header -->
    <header class="main-header">
      <span class="header-icon">🛒</span> Online Shopping
    </header>

    <!-- Section 1: Primitive Data Types -->
    <h2 class="section-title">Primitive Data Types</h2>
    <div id="primitiveContainer" class="result-box success-box">
      <div>String : <span id="stringVal" class="bold-text"></span></div>
      <div>Number : <span id="numberVal" class="bold-text"></span></div>
      <div>Boolean : <span id="booleanVal" class="bold-text"></span></div>
      <div>Undefined : <span id="undefinedVal" class="bold-text"></span></div>
      <div>Null : <span id="nullVal" class="bold-text"></span></div>
    </div>

    <!-- Section 2: Type Conversion -->
    <h2 class="section-title">Type Conversion</h2>
    <div class="form-container">
      <input
        type="text"
        id="typeInput"
        class="input-field"
        value="88"
        placeholder="Enter a value..."
        autocomplete="off"
      >
      <button id="btnCheckType" class="btn-validate">Check Type</button>

      <div id="typeResultBox" class="result-box success-box" style="display: none;">
        <div>Before Conversion : <span id="beforeType" class="bold-text"></span></div>
        <div>After Conversion : <span id="afterType" class="bold-text"></span></div>
      </div>
    </div>
  </div>
```

```

<!-- Section 3: Shopping Bill -->
<h2 class="section-title">Shopping Bill</h2>
<div class="form-container">
  <input
    type="number"
    id="price1Input"
    class="input-field"
    value="1000"
    placeholder="Price of Item 1"
  >
  <input
    type="number"
    id="price2Input"
    class="input-field"
    value="77899"
    placeholder="Price of Item 2"
  >
  <input
    type="number"
    id="qtyInput"
    class="input-field"
    value="988"
    placeholder="Quantity"
  >
  <select id="paymentDropdown" class="input-field select-field">
    <option value="Cash">Cash</option>
    <option value="Card">Card</option>
    <option value="UPI">UPI</option>
  </select>
  <button id="btnCalculateBill" class="btn-validate">Calculate Bill</button>

  <div id="billResultBox" class="result-box success-box" style="display: none;">
    <div>Subtotal : <span id="subtotalOutput" class="bold-text"></span></div>
    <div>Discount (<span id="discountPercent"></span>) : <span id="discountOutput"
class="bold-text"></span></div>
    <div>Total Payable : <span id="payableOutput" class="bold-text"></span></div>
  </div>
</div>
</div>

  <script src="script.js"></script>
</body>
</html>

```

### styles.css

```

body {
  font-family: Arial, sans-serif;
  background-color: #f8f9fa;
  color: #333;
  margin: 0;
  padding: 20px;
  display: flex;
  justify-content: center;
  align-items: flex-start;
  min-height: 100vh;
}

.app-container {
  width: 100%;
  max-width: 460px;
  background-color: #ffffff;
  border: 1px solid #ddd;
  border-radius: 6px;
  padding: 25px;
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.05);
  box-sizing: border-box;
  margin-top: 20px;
}

```

```

}

/* Main Header matching Online Shopping title */
.main-header {
  font-size: 28px;
  font-weight: bold;
  color: #0056b3;
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 8px;
  margin-bottom: 25px;
}

.header-icon {
  font-size: 26px;
}

/* Section Title with the blue underline border */
.section-title {
  color: #0056b3;
  font-size: 20px;
  font-weight: bold;
  border-bottom: 2px solid #0056b3;
  padding-bottom: 8px;
  margin-top: 25px;
  margin-bottom: 20px;
}

.form-container {
  display: flex;
  flex-direction: column;
  gap: 15px;
}

/* Full-width rounded input field */
.input-field {
  width: 100%;
  padding: 10px 12px;
  font-size: 15px;
  border: 1px solid #ccc;
  border-radius: 4px;
  box-sizing: border-box;
  outline: none;
}

.input-field:focus {
  border-color: #0056b3;
}

/* Select dropdown styling */
.select-field {
  background-color: #ffffff;
  cursor: pointer;
  height: 40px;
}

/* Full-width blue buttons matching the screenshot check/calculate buttons */
.btn-validate {
  width: 100%;
  padding: 12px;
  font-size: 15px;
  font-weight: bold;
  background-color: #0056b3;
  color: #ffffff;
  border: none;
  border-radius: 4px;
  cursor: pointer;
  box-sizing: border-box;
}

```

```

    text-align: center;
    transition: background-color 0.2s;
    margin-bottom: 5px;
}

.btn-validate:hover {
    background-color: #004085;
}

/* Result box using the light blue card styles */
.result-box {
    padding: 15px;
    border-radius: 4px;
    font-size: 14px;
    line-height: 1.6;
    box-sizing: border-box;
}

.result-box div {
    margin-bottom: 6px;
}

.result-box div:last-child {
    margin-bottom: 0;
}

.success-box {
    background-color: #ebf4fc;
    color: #333;
    border: 1px solid #d2e6f7;
}

.bold-text {
    font-weight: bold;
}

```

### script.js

```

document.addEventListener('DOMContentLoaded', () => {
    // 1. Primitive Data Types Implementation
    // Declare different primitive variable types
    const stringVar = "Adarsh";
    const numberVar = 1000;
    const booleanVar = true;
    const undefinedVar = undefined;
    const nullVar = null;

    // Display values on page load
    document.getElementById('stringVal').textContent = stringVar;
    document.getElementById('numberVal').textContent = numberVar;
    document.getElementById('booleanVal').textContent = String(booleanVar);
    document.getElementById('undefinedVal').textContent = String(undefinedVar);
    document.getElementById('nullVal').textContent = String(nullVar);

    // 2. Type Conversion Implementation
    const typeInput = document.getElementById('typeInput');
    const btnCheckType = document.getElementById('btnCheckType');
    const typeResultBox = document.getElementById('typeResultBox');
    const beforeType = document.getElementById('beforeType');
    const afterType = document.getElementById('afterType');

    btnCheckType.addEventListener('click', () => {
        const val = typeInput.value;

        // Input values are captured as string type by default from the DOM
        beforeType.textContent = typeof val;

        // Convert the string to a number explicitly using Number()
        const convertedVal = Number(val);
    });

```

```

        afterType.textContent = typeof convertedVal;

        // Show the results box
        typeResultBox.style.display = 'block';
    });

    // Run check type once automatically to display initial state
    btnCheckType.click();

    // 3. Shopping Bill Implementation
    const price1Input = document.getElementById('price1Input');
    const price2Input = document.getElementById('price2Input');
    const qtyInput = document.getElementById('qtyInput');
    const paymentDropdown = document.getElementById('paymentDropdown');
    const btnCalculateBill = document.getElementById('btnCalculateBill');
    const billResultBox = document.getElementById('billResultBox');

    const subtotalOutput = document.getElementById('subtotalOutput');
    const discountPercent = document.getElementById('discountPercent');
    const discountOutput = document.getElementById('discountOutput');
    const payableOutput = document.getElementById('payableOutput');

    btnCalculateBill.addEventListener('click', () => {
        // Read input values and convert them explicitly from string to number
        const price1 = parseFloat(price1Input.value) || 0;
        const price2 = parseFloat(price2Input.value) || 0;
        const qty = parseInt(qtyInput.value) || 0;
        const paymentMethod = paymentDropdown.value;

        // Calculate Subtotal using the formula: (Price 1 + Price 2) * Quantity
        const subtotal = (price1 + price2) * qty;

        // Determine discount percentage using conditional statements
        let discountRate = 0;
        if (paymentMethod === "Cash") {
            discountRate = 0.10; // 10% discount
        } else if (paymentMethod === "Card") {
            discountRate = 0.05; // 5% discount
        } else if (paymentMethod === "UPI") {
            discountRate = 0.15; // 15% discount
        }

        const discountAmount = subtotal * discountRate;
        const totalPayable = subtotal - discountAmount;

        // Format and display the results
        subtotalOutput.textContent = `₹${subtotal.toFixed(2)}`;
        discountPercent.textContent = `${discountRate * 100}%`;
        discountOutput.textContent = `₹${discountAmount.toFixed(2)}`;
        payableOutput.textContent = `₹${totalPayable.toFixed(2)}`;

        // Show the results box
        billResultBox.style.display = 'block';
    });
});

```

## 4. Output:

PRN: 24070521038 | Name: Vansh Mahant | File Path: /practical5/index.html — Page load (primitive types & type conversion)

## Online Shopping

### Primitive Data Types

String : **Adarsh**  
Number : **1000**  
Boolean : **true**  
Undefined : **undefined**  
Null : **null**

### Type Conversion

Check Type

Before Conversion : **string**  
After Conversion : **number**

### Shopping Bill

Calculate Bill

 **Online Shopping**

**Primitive Data Types**

String : **Adarsh**

Number : **1000**

Boolean : **true**

Undefined : **undefined**

Null : **null**

**Type Conversion**

88

Check Type

Before Conversion : **string**

After Conversion : **number**

**Shopping Bill**

1000

77899

988

Cash

Calculate Bill

Subtotal : ₹77952212.00

Discount (10%) : ₹7795221.20

Total Payable : ₹70156990.80

## 5. Result/Conclusion:

An Online Shopping page demonstrating JavaScript basics was successfully implemented using HTML, CSS and JavaScript. The program displays the five primitive data types (string, number, boolean, undefined, null), shows explicit type conversion from string to number using the `typeof` operator, and calculates a shopping bill (subtotal, discount and total payable) using conditional statements that apply a different discount rate depending on the payment method selected (Cash, Card, or UPI). This experiment reinforced the concepts of primitive data types, type conversion, and conditional logic in JavaScript.

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Upload the Soft copy as well as complete code on GitHub link is mandatory.

File name: Practical 2\_24070521038\_Vansh Mahant\_A\_A1